

Statistics formula sheet

Statistics You Can Use

Inkwell

Averages

Measure	How to find it
Mean	Sum of values / number of values
Median	Middle value when sorted; mean of the two middle values if the count is even
Mode	Most frequent value

Spread

Measure	How to find it
Range	Largest - smallest
Interquartile range	Upper quartile - lower quartile

Percentages

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

Percentage of a total = $\text{part} / \text{total} \times 100$

Choosing an average

- Outliers present: use the median
- Categories: use the mode
- Evenly spread numbers: use the mean

Reading charts critically

- ☐ Does the vertical axis start at zero?
- ☐ Are units and labels shown?
- ☐ Is the time range cherry-picked?
- ☐ Do pictures distort the size?
- ☐ Is correlation being sold as cause?

Worked example

Marks: 4, 7, 7, 8, 9, 13

Mean 8, median 7.5, mode 7, range 9.